

Back then, programs weren't transferred electronically. The machine code was punched into paper tape, a common storage medium of the time. Thompson would physically carry the paper tape to the PDP-7, feed it into the machine, and load the program manually.

Birth of UNIX

To escape the tedious development process, Thompson and Dennis Ritchie decided to implement their new system design directly on the PDP-7.

They built a self-supporting system — one that included an early version of the UNIX file system, a process subsystem, and a small set of utility programs. For the first time, they didn't need GECOS anymore.

Their creation was independent, elegant, and efficient.

When the time came to name it, their colleague Brian Kernighan jokingly suggested 'UNIX' — a pun on 'Multics', since it was a simpler, 'uni-tasking' version of the overcomplicated Multics.

First steps into the real world

The first version of UNIX was experimental but promising. Its first real application was a text-processing system for the Bell Labs' patent department — a project that demonstrated UNIX's practicality and power.

By 1971, UNIX was ported to the PDP-11, a more powerful computer, marking the beginning of its evolution.

Even though the system was incredibly lightweight with:

- 16KB memory for the OS
 - 8KB for user programs
 - 512KB of total disk space
 - 64KB maximum file size
- ...it was fast, reliable, and efficient.

The birth of the C language

Ken Thompson initially wanted to build a Fortran compiler, but instead created a simpler language called B, inspired by BCPL (Basic Combined Programming Language). However, B was interpreted, executing code line by

line, making it too slow.

To solve this, Dennis Ritchie developed a new language called C, which allowed compilation to machine code, supported data types, and enabled structured programming.

In 1973, UNIX was rewritten in C — a revolutionary move at the time, since operating systems were almost always written in assembly language.

This decision made UNIX portable, modifiable, and easy to improve.

Spreading through Bell Labs and academia

After UNIX proved successful, Bell Labs installed it widely — about 25 internal installations. To manage it, they formed the UNIX System Group. However, due to a 1956 US government consent decree, AT&T (Bell's parent company) was legally prohibited from selling computer products commercially. Instead, Bell Labs shared UNIX freely with universities and research institutions for educational purposes. This generosity made UNIX's reputation grow organically.

In 1974, Ritchie and Thompson published their now-legendary paper, 'The UNIX Time-Sharing System', in the *Communications of the ACM*, the monthly journal of the Association for Computing Machinery, introducing UNIX to the world.

By 1977, there were over 500 UNIX installations worldwide, including 125 universities using it for teaching, research, and software development. That same year, the Interactive Systems Corporation became the first value-added reseller (VAR) of UNIX, enhancing it for office automation.

UNIX's portability was further proven when it was successfully ported from PDP computers to the Interdata 8/32, a completely different architecture.

Fragmentation and standardisation

Between 1977 and 1982, as microprocessors became popular,

various companies began porting and modifying UNIX for their own hardware. UNIX's open and modular design encouraged developers to extend and customise it, but this also led to many incompatible variants.

To bring order, Bell Labs consolidated its versions into a unified commercial release — UNIX System III, later refined into UNIX System V in January 1983. Meanwhile, researchers at the University of California, Berkeley, developed their own improved version — the Berkeley Software Distribution (BSD).

By the time 4.3BSD was released, UNIX had included advanced features such as improved networking, virtual memory, and tools that would later influence the internet's foundation.

A global phenomenon

By 1984, UNIX had achieved over 100,000 installations worldwide. It ran on everything from microprocessors to mainframes, supporting multiple hardware platforms and manufacturers — a feat unheard of at the time.

What began as an experiment to run a space game had become a global computing standard, shaping decades of innovation that followed.

The legacy of UNIX

From a tiny PDP-7 machine to the world's data centres and smartphones, the UNIX philosophy — simplicity, portability, and modularity — continues to define how we design software today.

Its descendants — Linux, BSD, macOS, Android, and more — still carry its DNA.

The story of UNIX is not just about technology; it's a testament to curiosity, creativity, and the pursuit of elegant simplicity. What started as a personal project in a Bell Labs lab forever changed the world of computing. 

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